Balagh — Project Cost Analysis

Full cost of running the Balagh incident-reporting platform in production: app-store fees, cloud/compute, database, and maps — modeled across 100 → 200,000 monthly active users.

Cloud: AWS vs DigitalOcean | Maps: Mapbox vs Google vs Self-hosted | iOS + Android stores

Prepared 18 June 2026 · All figures USD · List prices, no reserved/committed-use discounts · Pricing live-verified June 2026 (see Sources).

1. Executive summary

Balagh is a lightweight, privacy-first, **text-only** safety app: two small Node.js (Fastify) services, one PostgreSQL + PostGIS database, a static web dashboard, and a React Native mobile app using the **Mapbox** Maps SDK. There is no media, no object storage, and no message queue, so the infrastructure footprint is small and the only usage-scaled cost driver is the **maps vendor**.

Recommendation: Keep **Mapbox** and deploy on **DigitalOcean** as a single shared stack. It matches the app as already built, is **free up to 25,000 monthly active users**, is more privacy-aligned than Google, and DigitalOcean is markedly cheaper and simpler than AWS at every tier modeled. Revisit **self-hosted tiles** as you approach 25k MAU (where Mapbox starts billing); move to an **AWS high-availability** design only when reliability or compliance demands it.

Users (MAU)	Total / month*	Total / year (yr 1)*	What dominates the bill
100	~$12	~$268	One small server; maps free
1,000	~$27	~$448	Server + managed DB; maps free
10,000	~$54	~$772	Bigger server + DB; maps still free
100,000	~$450	~$5,524	Mapbox MAU fees ($300/mo) overtake cloud
200,000	**~$890**	**~$10,804**	**Mapbox ($640/mo) is the majority of cost**

*Recommended stack: DigitalOcean (single shared) + Mapbox + app-store fees. Year-1 includes Apple $99 + Google $25 one-time. Full breakdown in §7.

2. What Balagh is (and why it's cheap to run)

From `mobile/MobileFrontendSpecs.md` and the repository:

- **Mobile app:** bare React Native 0.85.3, Mapbox Maps SDK v11 (`@rnmapbox/maps`), local AsyncStorage only. No camera/microphone/photos, **no Firebase/FCM** (push deferred). Anonymous device identity, no accounts.
- **Backend:** two Fastify (Node 22) services — a public mobile API and an operator API — sharing one **PostgreSQL 16 + PostGIS 3.4** database. Real-time via in-process WebSocket (no Redis / queue / broker).
- **Web dashboard:** React + Vite static SPA using MapLibre GL with free CARTO basemap tiles — hostable as static files (≈ free).
- **No object storage, no media, text-only payloads** (reports/comments ≤ 280 chars). Map tiles are served from the maps vendor's CDN, not our bandwidth — unless we self-host. This keeps egress and compute small.

Cost consequence: compute and bandwidth stay modest as users grow; the single steeply-scaling line item is the **maps vendor's per-MAU fee**.

3. App-store costs (iOS + Android)

Item	Apple App Store	Google Play
Account / registration	**$99 / year** (Apple Developer Program, recurring)	**$25 one-time** (no renewal)
Enterprise / in-house only	$299/yr (private distribution — *not needed*)	n/a
Certificates, signing, review, beta (TestFlight / internal testing)	Included	Included
Store commission (paid apps / in-app purchases / subscriptions)	**15–30% — but Balagh is free with no in-app purchases → $0 commission**	
Org account prerequisites	D-U-N-S number (free)	Business verification (free)
Effective recurring store cost	**$99 / year**	**$0 / year ($25 once, year 1)**

Net store cost = $99/yr + $25 first-year one-time = $124 in year 1, then $99/yr thereafter — flat regardless of user count, because the app is free and takes no payments.

4. Maps service comparison

The mobile app currently uses the **Mapbox Maps SDK**, billed by **monthly active users (MAU)**. Two alternatives: Google Maps (free for mobile map display) and self-hosting our own vector tiles (MapLibre + Protomaps/OpenMapTiles).

4.1 Monthly maps cost by scale

Users (MAU)	Mapbox	Google Maps	Self-hosted
100	$0	$0	~$0
1,000	$0	$0	~$0
10,000	$0	$0	~$5
100,000	$300	$0	~$30 + egress
200,000	**$640**	**$0**	**~$60 + egress**

Mapbox rate card: free ≤ 25,000 MAU; 25,001–125,000 = $4.00/1,000; 125,001–250,000 = $3.20/1,000; 250,001+ = $2.40/1,000. **100k** = (100,000−25,000)/1,000 × $4 = **$300/mo**. **200k** = 100,000×$4/1k ($400) + 75,000×$3.20/1k ($240) = **$640/mo**.

Google Maps: the Maps SDK for Android & iOS (dynamic map display) is **free with unlimited usage**. Other SKUs (Geocoding, Places) get 10,000 free events/month then ~$2–7/1,000 — Balagh barely needs them.

4.2 Trade-offs

Option	Cost	Privacy fit	Engineering effort	Verdict
Mapbox (current)	Free to 25k MAU, then per-MAU	**Good — no Google dependency**	None (already built: privacy circles, Arabic labels)	**Recommended now**
Google Maps	**$0 for map display**	**Weak — routes users through Google, against app ethos**	High — rewrite away from Mapbox-specific privacy-circle / Arabic-label code	**Cheapest, but ethos conflict**
Self-hosted (MapLibre + own tiles)	Low ongoing (server + egress); high one-time	**Best — fully in-house, no third-party tracking**	High one-time — tile pipeline, rebuild style, re-do Arabic labels	**Best at scale (≥25k MAU)**

Israel is a tiny map extract, so self-hosted tiles are cheap to store/serve; the real cost is one-time engineering. The crossover where it pays back is when Mapbox MAU fees become material — i.e. around and beyond **25,000 MAU**.

5. Cloud deployment comparison (AWS vs DigitalOcean)

Scenario: **single shared stack** — both Node services + PostgreSQL/PostGIS in one environment; web dashboard served as static files. Sizing grows with load.

Users (MAU)	DigitalOcean — components	DO / mo	AWS — components	AWS / mo
100	1× Droplet 2 GB, Postgres on-box	~$12	Lightsail $10–12 (all-in)	~$12
1,000	Droplet 2 GB + Managed PG 1 GB ($15)	~$27	Lightsail + RDS db.t4g.micro ($22)	~$34
10,000	Droplet 4 GB/2 vCPU ($24) + Managed PG 2 GB ($30)	~$54	EC2 t4g.medium + RDS db.t4g.small ($51) + ALB	~$95
100,000	2× app Droplets + PG HA (primary+standby $60) + LB ($12)	~$150	HA EC2 (2×) + RDS Multi-AZ + ALB + transfer	~$330
200,000	**3× app Droplets + larger PG HA + LB + Spaces**	**~$250**	**EC2 fleet + RDS Multi-AZ + ALB + transfer**	**~$520**

DigitalOcean

- Droplets $6 (1GB) / $12 (2GB) / $24 (4GB) / $48 (8GB)
- Managed PostgreSQL from $15/mo; HA from $60/mo
- Load balancer $12/mo; Spaces $5/mo
- Generous pooled data transfer → egress ≈ $0 at our scale
- **Simpler, cheaper, predictable**

AWS

- Lightsail $5–$44 all-in; EC2 t4g.small ≈ $12 + EBS
- RDS db.t4g.micro $21.90/mo; db.t4g.small $51.10/mo (×~2 for Multi-AZ)
- ALB ≈ $16–22/mo; data transfer $0.09/GB after 100 GB free
- More services, steeper learning curve
- **Pick when HA / compliance / scale demand it**

Bandwidth: Balagh is text-only JSON + small WebSocket keepalives; map tiles come from the maps CDN. Egress is modest (well within DO's included transfer; tens of GB/mo on AWS at 100k). **Self-hosting map tiles** would shift tile bandwidth onto our bill — the main reason to budget egress/CDN if going that route.

6. Total cost of ownership — all scenarios

Year-1 totals = (cloud × 12) + (maps × 12) + Apple $99 + Google $25 one-time. Subsequent years drop the $25.

6.1 DigitalOcean + Mapbox (recommended)

Users	Cloud	Maps	Stores (yr)	Total / month	Total / year
100	$12	**$0**	$124	~$12	~$268
1,000	$27	**$0**	$99	~$27	~$448
10,000	$54	**$0**	$99	~$54	~$772
100,000	$150	**$300**	$99	~$450	~$5,524
200,000	**$250**	**$640**	**$99**	**~$890**	**~$10,804**

6.2 DigitalOcean + Google Maps (cheapest, ethos caveat)

Users	Cloud	Maps	Stores (yr)	Total / month	Total / year
100	$12	**$0**	$124	~$12	~$268
1,000	$27	**$0**	$99	~$27	~$448
10,000	$54	**$0**	$99	~$54	~$772
100,000	$150	**$0**	$99	~$150	~$1,899
200,000	**$250**	**$0**	**$99**	**~$250**	**~$3,099**

6.3 AWS + Mapbox (production / HA-leaning)

Users	Cloud	Maps	Stores (yr)	Total / month	Total / year
100	$12	**$0**	$124	~$12	~$268
1,000	$34	**$0**	$99	~$34	~$507
10,000	$95	**$0**	$99	~$95	~$1,239
100,000	$330	**$300**	$99	~$630	~$7,659
200,000	**$520**	**$640**	**$99**	**~$1,160**	**~$13,999**

7. Grand total — the bottom line

Recommended path (**DigitalOcean + Mapbox**), all costs included (cloud + maps + iOS + Android):

Monthly active users	Total per month	Total per year (year 1)	Total per year (ongoing)
100	~$12	~$268	~$243
1,000	~$27	~$448	~$423
10,000	~$54	~$772	~$747
100,000	~$450	~$5,524	~$5,499
200,000	**~$890**	**~$10,804**	**~$10,779**

Reading the table: up to ~10,000 users the whole platform runs for the price of a couple of coffees a month — maps are free and one small server suffices. Cost only becomes significant past 100k users, and there it is dominated by the Mapbox per-MAU fee, not infrastructure — which is exactly when self-hosted tiles (or, with caveats, Google's free SDK) become worth the engineering investment.

8. Assumptions & disclaimer

- **Moderate engagement:** user count = MAU; ~10 app opens + map views per user per month, a few small JSON fetches each, occasional report. Text-only, no media.
- **Single shared stack;** web dashboard served as static (≈ free).
- USD list prices, no reserved/committed-use discounts; one-time engineering labour for a maps migration is excluded (called out qualitatively in §4).
- Cloud sizing is an estimate; real load testing may shift a tier up or down. Prices were live-verified in June 2026 and **should be re-checked before budgeting.**

9. Sources (verified June 2026)

- Mapbox pricing — https://www.mapbox.com/pricing
- Google Maps Platform pricing — https://mapsplatform.google.com/pricing/
- DigitalOcean PostgreSQL — https://docs.digitalocean.com/products/databases/postgresql/details/pricing/
- DigitalOcean Droplets — https://www.digitalocean.com/pricing/droplets
- AWS RDS instance pricing — economize.cloud / aws.amazon.com (db.t4g.micro $21.90/mo, db.t4g.small $51.10/mo)
- AWS Lightsail — https://aws.amazon.com/lightsail/pricing/
- Apple Developer Program $99/yr; Google Play $25 one-time — developer.apple.com / play.google.com/console